Ajoy Sarmah

Mechanical Engineer

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SUMMARY

- Mechanical Engineer with an M.Tech in Thermal and Fluid Engineering and a published research paper on supercritical CO_2 power cycles.
- Skilled in ANSYS, CATIA, SolidWorks, and FLOWNEX with experience in oilfield project coordination, mechanical systems, and technical analysis.
- Seeking to leverage technical expertise in a reputable engineering organization.

SKILLS

Engineering Software | FLOWNEX | ANSYS | CATIA | SOLID WORKS

Programming | MATLAB | C++ | C

Core Mechanical Areas | Thermodynamics | Fluid Mechanics | Heat Transfer | Energy Systems

Other Skills | MS OFFICE (Word, Excel, Power Point, Outlook) | LEADERSHIP | TIME MANAGEMENT

WORK EXPERIENCE

Project Coordinator — Oct 2024 - Present

Gtc Oilfield Services Limited , Duliajan

- Coordinate with multiple departments to ensure smooth execution of oilfield drilling and project operations.
- Manage allocation of manpower, equipment, and materials to maintain uninterrupted rig activities.
- Oversee procurement and inventory tracking of essential operational supplies.
- Act as a liaison between project managers, drilling teams, and clients for operational coordination.
- Maintain records of project progress, permits, and compliance documentation.

Intern — Jun 2018 - Jun 2018

Assam Gas Company Limited , Duliajan

- Gained exposure to compressors, engines, and gas flow management systems used for distribution to consumers.

Intern — May 2018 - May 2018

Oil India Limited , Duliajan

- Gained hands-on experience with the operation of drilling mechanisms.
- Gained knowledge on various engines and pumps used in the extraction of crude oil.

Intern — Dec 2016 - Jan 2017

Tata Motors

- Acquired practical knowledge of automobile components, engine systems, and vehicle mechanical assemblies.

EDUCATION

M. Tech: Tezpur University | **CGPA: 7.87** | **Nov 2022 - May 2024**

Mechanical Engineering

- Presented research work at ICFTES 2024 hosted by NIT Calicut on supercritical CO_2 cycle analysis.
- Attended workshop on "Adsorption on Carbon Dioxide Capture, Green Refrigeration and Energy Storage" conducted by IIT Dhanbad.
- Completed workshop on "Elements of Style in Technical Communication" conducted by Tezpur University.

B. Tech: Kalinga Institute Of Industrial Technology | **CGPA: 8.20** | **Jun 2015 - Jun 2019**

Mechanical Engineering

- Developed strong fundamentals in thermodynamics, fluid mechanics, heat transfer, machine design, and manufacturing.
- Completed an experimental project on the Solar Organic Rankine Cycle (SORC) for small-scale power generation.
- Active member of Kalinga Institute of Social Science initiatives and university activities.

PROJECTS

Supercritical CO_2 Power Cycle Design | **Aug 2023 - Jun 2024**

- Designed a supercritical CO_2 power cycle integrated with a sodium-cooled fast reactor using FLOWNEX software.
- Conducted energy, exergy, and exergoeconomic analyses for system performance evaluation.

Solar Organic Rankine Cycle (SORC) Prototype | **Jun 2018 - Jun 2019**

- Developed an experimental solar ORC prototype using butanol as working fluid.
- Evaluated system feasibility for power generation in Indian climatic conditions.

PUBLICATIONS

- Published the research article "Proposal for a Combined Supercritical CO_2 and Organic Rankine Cycle: System Design and Exergoeconomic Analysis" in the journal *Thermal Science and Engineering Progress* (May 2025).

ADDITIONAL INFORMATION

- **Languages**: English | Hindi | Assamese
- **Interests**: Table Tennis, Chess, Volleyball, Music (Tabla)